

Ved Patel

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

Aug 2023 – May 2025

- Relevant skills: Web Technologies, Deep Learning, Multimodal techniques for different Modalities, Security Systems, Database Management, Analysis of algorithms, Advanced Data Storage

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Computer Science

Jun 2019 – May 2023

- Relevant skills: Operating Systems, Computer Networks, Financial Services, Computer Architecture, Cloud Computing, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, Dart, C++, Java, HTML, CSS, JavaScript

Framework: Flutter, Bootstrap, Flask, TensorFlow, ReactJs, NodeJs, Express

Libraries: Keras, Seaborn, Matplotlib, Pandas, Numpy, PennyLane, Python coco, OpenCV

Databases: Firebase, SQL, Mongo DB, PostgreSQL, DuckDB

Deployment: AWS(EC2, Lambda, S3, IAM, SageMaker), Docker, Kubernetes, Google Cloud, Azure

Methodologies: Agile, Scrum, Gitflow, Object Oriented Design, Microservices architecture

EXPERIENCE

AI Software Engineer

May 2024 – Nov 2024

Kintsugi Global Inc.

Los Angeles, CA

- Integrated an AI avatar for speech-to-text and text-to-speech functionalities leveraging **Hubert**, **PyTorch**, and the Speech Recognition JS model also deployed **AWS SageMaker** to run these large models, boosting runtime efficiency by 150%
- Constructed responsive frontend designs leveraging **React**, combining **Tailwind CSS** for streamlined styling and increased user experience, while employing lazy loading to increase efficiently by 25% for loading 3D avatars
- Deployed a secure backend with **Express.js (Node.js)** for efficient data transfer, managing functionalities store and manga, while accomplishing protections against screenshots and unauthorized access

Software Engineer

Dec 2023 – Apr 2024

Sequel

Los Angeles, CA

- Utilized state management and component-based architecture in **React** to enhance readability and simplify debugging, while optimizing performance through React hooks for efficient state handling
- Performed in-depth research on top job board websites, utilizing **Power BI** to analyze user engagement metrics and features to uncover improvement opportunities
- Constructed an automation bot for marketing team employing **Google Sheets App Script (JavaScript)**, reducing workload by more than 50% and streamlining processes

Application Developer

Jan 2022 – Mar 2022

Two loops technologies

Chennai, India

- Engineered and launched inaugural app, demonstrating frontend expertise in **Flutter and Firebase**, and integrating **OneSignal** for push notifications, increasing user retention by 30% and enhancing engagement insights
- Constructed audio functionality in Flutter, including saving audio files in Firebase Firestore, storing IDs in real-time data, and visualising audio waves based on amplitude of input audio, enriching user interaction within app

PROJECTS | PUBLICATIONS

PostgreSQL & DuckDB Optimization | C++

Jan 2025 – Apr 2025

- Implemented a FIFO queue-based approach in PostgreSQL to improve memory efficiency and enhanced DuckDB's performance by optimizing concurrency and query execution for faster, more efficient data handling

Stock Predication Using CandleStick Patterns | Tensorflow, YOLOv9

Aug 2024 – Dec 2024

- Developed a stock prediction model combining **YOLOv9** for candlestick pattern detection and **LSTM** for time-series forecasting to analyze mid-term trends

Stock Simulation | Java, MERN

Jan 2024 – Apr 2024

- Developed a dynamic Android mobile application for real-time stock tracking, enabling users to view stock price changes, access the latest news, and perform buy/sell transactions

Multimodal Laughter Detection | PyTorch, Hubert, Marlin

Jan 2024 – Apr 2024

- Created a multimodal laughter detection system operating **Hubert** for audio and **Marlin** for video feature extraction and achieved 87.08% accuracy using a late fusion approach, improving detection accuracy compared to individual modalities

Employing Machine Learning to Identify Waste Characteristics | pylab, python coco tools

Aug 2022- Jan 2023

- Developed a waste classification system using **numpy**, **pandas**, **matplotlib**, **seaborn**, **python coco tools**, and **pylab** to automate waste categorization and provide statistical insights for more effective waste management | [Research Paper](#)